SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I G. Amrutha (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

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Short Answer Test

CO1-AT3 - Cloud Computing

CSA1520

1) Define cloud computing and explain any four characteristics that make it different from traditional onpremise computing.

2) Trace the evolution of cloud computing from hardware evaluation to internet software evaluation, and show how these changes enabled modern cloud platforms.

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

4) Discuss the role of data centers in delivering cloud services.

5) Differentiate IaaS, PaaS, SaaS and XaaS with one suitable real-world example for each model.

1A Cloud Computing:

Cloud computing is the delivery Computing the resources such as storage, networking, database, and software over the internet.

Characteristics:

1) on-demand self service: users can create and use resources whenever the required without manual interaction.

2) Resource pooling: It can computing the resources as shared among multiple users using the virtualization.

3) Broad network access: the servies can be accessed through the internet using laptops, mobiles, computers or other devices.

4) Rapid elasticity: The resources can be quickly increased or decreased according to the demand.

Different from traditional onpremise Computing:

Traditional onpremise computing requires organizations to, purchase, maintain and manage their own hardware and software.

The evolution of cloud computing occured through several storage.

Hardware evolution → virtualization → internet → web services → cloud computing.

⇒ main frames: Centralized computing with many users sharing one powerful computer.

⇒ High speed Networking: Connected computers and enabled resource sharing.

⇒ Internet and web: Applications and resources became accessible globally.

web services and APIs: Software could be delivered over networks.

Cloud platforms: providers can combined the virtualization, data centers, networking, automation, and internet access to provide IaaS, PaaS, SaaS and other services.

Therefore, the improvements in hardware networking, virtualization, and internet software enabled modern cloud platforms.

3A: Server Virtualization:

It divides one physical server into multiple virtual machines (VMs). Each VM can run its own operating system and applications

Role in cloud Computing:

=) Improves hardware utilization.

=) Allows multiple customers to share physical infrastructure.

=) provides isolation between users.

=) Supports scalability and resource management.

Virtualization	Cloud Computing
Technology that creates virtual resources from physical resources.	Service model that delivers computing resources over network.
Mainly focuses on efficient resource utilization.	Focuses on delivering resources as services.
Can exist without cloud computing.	Commonly uses virtualization as an enabling technology.

Example:

VMware virtualization can create VMs, while AWS, while AWS provides cloud services using technologies including virtualization.

Data Centers :

A Data centers is a facility containing servers, storage systems, networking equipment, and power systems.

Roles of data centers :

=) Host cloud applications and services.
=) Storage and process large amount of data.
=) Provide networking and internet connectivity
=) Provide security and controlled physical access.
=) It supports scalability by adding computing resources when demand increases.

Therefore, the data centres from the physical foundation of Cloud Computing.

Sr.	Model	Meaning	What it provides...

1) IaaS :
=> Infrastructure as a service.
=> Virtual machines, Storage, networks.
=> Ex: Amazon EC2.

2) PaaS :
=> Platform as a service.
=> Development platform, runtime, tools and databases.
=> Ex: Google App engine.

3) SaaS :
=> Software as service
=> Ready-to use software through the internet.
=> Ex: Google workspace.

4) XaaS :
=> Everything/Anything as a service.
=> Various IT capabilities delivered as service.
=> Ex: AWS Lambda

IaaS → Infrastructure → PaaS → platform → SaaS → Software → XaaS → Almost any IT service delivered as a service.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

